compute • calcul CANADA

TOWN HALL MEETINGS

May 2010

INVITATION

As a user of high performance computing facilities, you are invited to an important Town Hall meeting to discuss the future of high performance computing in Canada. We would like your input on a number of key questions and issues related to resource allocation, computing resources, user support and the organization of Compute Canada. Please join us and take advantage of this opportunity to express your point of view!

CONTEXT

The purpose of the town hall meetings is to engage researchers in a discussion of the strategic direction for Compute Canada and the continuing development of the national HPC platform. We would like to hear your thoughts and ideas with respect to the organization of Compute Canada, as well as the distribution of computing resources, and what can be done to keep Canada competitive in leading-edge HPC and give Canadian researchers a competitive advantage through HPC.

The questions and issues provided in this document are intended to focus our discussions but are in no way meant to limit them. Your identification and discussion of additional issues of importance to academic researchers in the context of Compute Canada are most welcome.

There will be a note-taker for each town hall meeting and a short summary report will be posted on the Compute Canada web site.

The town hall meetings will provide valuable input to the development of Compute Canada's strategic plan and our application for funding in the next national platform call for proposals.

You may wish to review the Compute Canada submission to the Canada Foundation for Innovation for the ***Mid-Term Review***. A link to the document may be found on the home page of the Compute Canada web site www.computecanada.org .

About Compute Canada

Compute Canada is leading the creation of a powerful national HPC platform for research. This national platform integrates High Performance Computing (HPC) resources at seven partner consortia across the country to create a dynamic computational resource. Compute Canada integrates high-performance computers, data resources and tools, and academic research facilities around the country. These integrated resources represent close to a petaflop of computing capability and online and long term storage with rapid access and retrieval over Canada's national, provincial and territorial high-performance networks. Working in collaboration, Compute Canada and the university-based regional HPC consortia provide for overall architecture and planning, software integration, operations and management, and coordination of user support for the national HPC platform. As a national organization, Compute Canada coordinates and promotes the use of HPC in Canadian research and works to ensure that Canadian researchers have the computational facilities and expert services necessary to advance scientific knowledge and innovation.

TOWNHALL QUESTIONS and ISSUES

Resource Allocation

1. Is the current resource allocation process working for you? Do you have any questions about the NRAC or LRAC processes?

Computing Resources

2. Compute Canada must reduce the number of data centres. What mechanisms/criteria should be used to determine the location of future (NPF-2) HPC equipment? Should external reviews be conducted? In what way, positively or negatively, would such a reduction impact your research?

3. Could Compute Canada and its partner consortia be organized in a way that would better address the needs of researchers to access machines across the country in an efficient and timely manner? Do researchers need to be geographically close to machines? Or is the critical factor having technical support geographically close?

4. How important is it that Compute Canada provide the full capability HPC ecosystem to the research community (even if you might not use it yourself)?
    - Should there be a Tier 1 component in the next NPF?
    - Should Compute Canada buy computing from the cloud?
    - What specialized services, software, infrastructure or resources should be available?

User Support

5. Are there circumstances under which researchers would be willing to pay for services (extensive technical consultation/unique software/other)?

6. Are you receiving the technical support you require to optimize your research?

7. Do you think your institution does enough for HPC?

8. Should, and if so, how should Compute Canada address the security and confidentiality issues of various research groups; i.e. medical (data/histories/HIPPA and 21 CFR part 11 compliance), security organizations (RCMP/CSIS), commercial data?

Organization

9. What changes to the structure of Compute Canada are necessary to ensure that decisions and outcomes are in the best interests of researchers across the country? In particular, how should the roles of regional consortia evolve in relation with Compute Canada?

10. There is a single consortium for the western provinces and a single consortium for the Atlantic provinces. What would you see as the benefits or potential drawbacks of a single consortium for each of Ontario and Quebec? What organizational model would ensure the best support for researchers?

11. Would HPC in Canada benefit if Compute Canada represented all HPC with members from government departments and the private sector? Would the "academic" voice be stronger with additional partners?